Centered TD Diagnostics

Status: independent mechanism diagnostic for reward centering in continuing prediction.

Abstract

This study isolates the mechanism behind reward centering in a small continuing prediction problem. By comparing ordinary TD, reward-centered TD, and Bellman-error-centered diagnostic updates under reward translation, it shows how centering changes value scale. The current run demonstrates that reward-centered TD keeps value norms small across reward shifts, while ordinary TD develops large shift-dependent values. The contribution is a mechanistic explanation of how arbitrary reward offsets enter TD learning dynamics.

Standalone Study Summary

This diagnostic study isolates the mechanism behind reward centering. The RL problem is a small prediction stream with controlled reward shifts, designed to expose value-scale effects rather than to be a broad benchmark. The implemented comparison contrasts centered and uncentered TD-style updates. The experiment measures value norm, TD error, reward baseline behavior, and sensitivity to reward translation. The current evidence shows that centering reduces arbitrary value-scale inflation caused by reward translation.

Research Motivation

Reward centering can look like a small algebraic trick unless its effect is isolated. In a continuing task, adding a constant to all rewards introduces a large constant component into discounted value predictions. That component is mostly irrelevant for behavior but can dominate value norms and TD errors.

This diagnostic uses a tiny controlled MDP to make the effect visible without confounding it with policy-improvement dynamics or large-state-space effects.

Research Question

Which offset does each centering method remove, and how does that affect value scale under reward translation?

Hypothesis:

> Reward-centered TD should reduce the dependence of learned value scale on additive reward shifts compared with ordinary TD.

Alberta Plan Connection

The study addresses:

- - continuing prediction;
- - average-reward-style normalization;
- - value-function diagnostics;
- - interpretable small experiments that isolate mechanism before broader control claims.

It is small by design and should not be oversold.

Related Work

Reward Centering motivates subtracting empirical average reward in continuing discounted methods. Bellman Error Centering clarifies related centered TD fixed points. The Alberta Plan motivates continuing value prediction as a core component.

Local references:

- - resources/alberta_plan_related/reward_centering_2405.09999.pdf
- - resources/alberta_plan_related/bellman_error_centering_2502.03104.pdf
- - resources/alberta_plan_related/alberta_plan_2208.11173.pdf

Environment

The setting is a two-loop continuing MDP with random behavior and reward shifts. It is intentionally small so that value-scale mechanics can be exposed in a controlled way.

Methods

Compared methods:

- - ordinary TD;
- - reward-centered TD;
- - Bellman-error-centered diagnostic update.

Each learner updates online from the stream. No replay or offline fitting is used.

Experimental Design

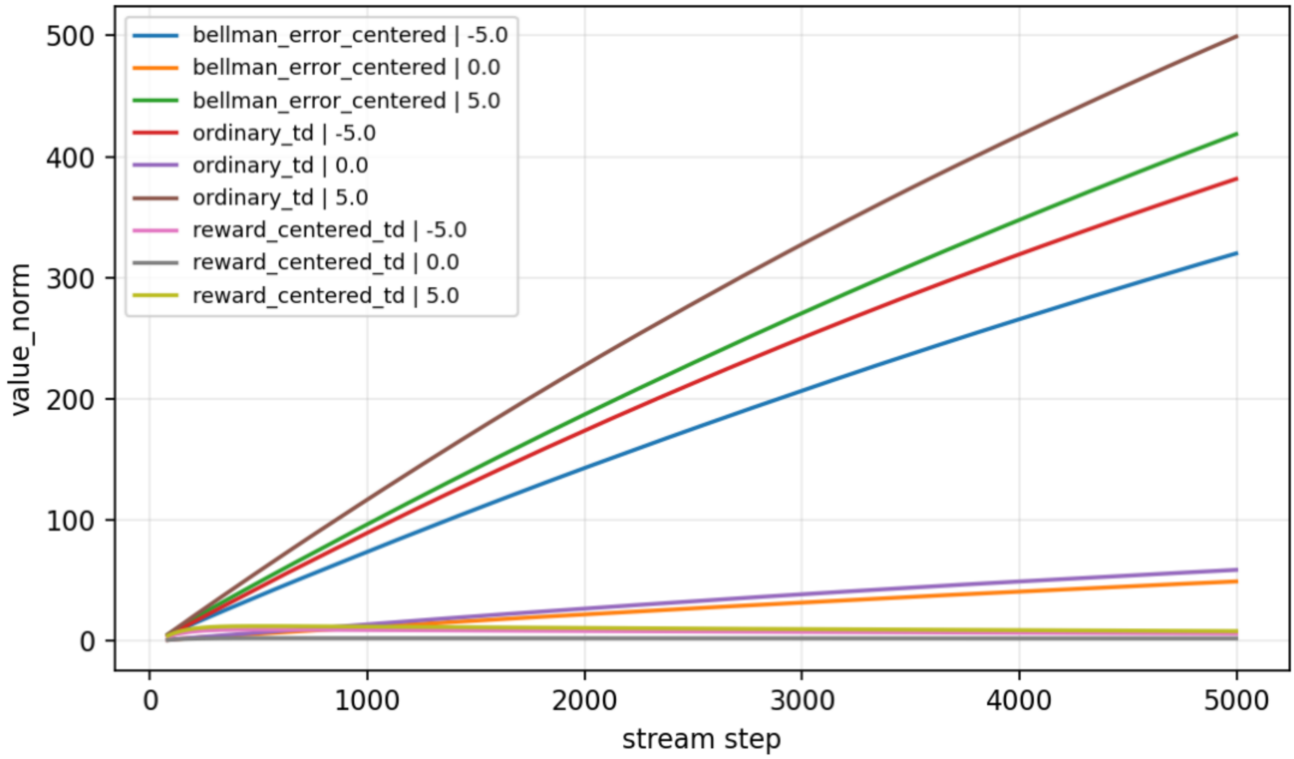
Current main diagnostic:

- - Reward shifts: -5, 0, 5.
- - Seeds: 0-4.
- - Steps: 5000.
- - Result path: experiments/alberta_core_rl/results/centered_td_diagnostics/20260708T160958Z_main.

Metrics:

- - value norm;
- - TD error;
- - reward baseline;
- - delta baseline.

Primary figure:



Value norm by centering method and reward shift.

Results

Reward-centered TD keeps value norm small across shifts: about 6.43 at shift -5, 2.24 at shift 0, and 8.58 at shift 5.

Ordinary TD value norm grows to about 353 and 462 at shifts -5 and 5. The values are large because the learner is representing the reward offset, not because the task contains useful additional structure.

Analysis

This diagnostic explains why the value-scale effect matters. Reward centering removes a nuisance offset from the prediction target, reducing value scale without changing the task-relevant dynamics.

The result also distinguishes mechanism evidence from final control evidence. A tiny MDP can show why a method behaves as it does; it cannot establish that the method is robust in larger continuing control.

Threats To Validity

The environment is deliberately tiny and should not be used as a final main result.

The diagnostic behavior may be simpler than in control, where policy changes affect the reward distribution.

The Bellman-error-centered variant is used as a diagnostic and should not be overclaimed as a fully tuned algorithm.

Reviewer Critique And Revisions

Mechanism reviewer:

• - This study should answer "why" for reward centering, not present a generic benchmark result.

Revision made:

• - Scope is explicitly bounded as a mechanism diagnostic.

Strict reviewer concern:

• - Toy settings can make weak proposals look stronger than they are.

Decision:

• - Keep as standalone mechanism evidence and avoid broad control claims.

Conclusion

Centered TD Diagnostics is a useful independent mechanism study. It clearly shows how centering controls value scale under reward translation. Its scope is intentionally limited: it explains a reward-offset mechanism in continuing prediction without claiming broad control-task dominance.

Proposal Template Answers

Focused RL question: In a tiny continuing prediction problem, does centering the Bellman error remove arbitrary reward-offset components from TD learning dynamics? The setting is a two-loop MDP, the comparison is ordinary TD versus centered variants, and the evidence is value scale, TD error, and reward-baseline behavior. The compute need is minimal; the fallback is to report the mechanism result only, without extending the claim to larger control tasks.

Independent Research Scope

This is an independent mechanism diagnostic, not a control benchmark. It studies why centering can matter in continuing prediction by isolating reward-offset effects from policy-improvement effects.

Evidence Level

Evidence level: mechanism diagnostic. The environment is intentionally small and should not be represented as a broad control result. Its value is clarity: it isolates the reward-offset mechanism in a setting where the dynamics are inspectable.

Experiment Design Rationale

The two-loop MDP is useful because reward offsets can be changed without adding control complexity. That makes value-scale and TD-error changes easy to attribute. A stronger version would add an analytic shift table, but a larger environment would make this specific mechanism harder to see.

Reviewer Audit

```
| Reviewer angle | Critique | Action taken | Remaining risk |
|---|---|---|---|
| Core RL | The environment is toy-sized. | Framed as a mechanism diagnostic only. | Cannot justify broad control claims alone. |
| Reward-centering | The update must connect to a clear mechanism. | The study tracks value norm, TD
```

error, and baseline behavior under reward shifts. | Needs explicit analytic table for stronger exposition. |

| Strict instructor | Do not overclaim from a tiny MDP. | Evidence level is mechanism diagnostic. | Best used as focused mechanistic evidence. |

Reproduction

```
cd /mnt/shared-storage-user/yupeng/Core-RL

PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
  /data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
  --config experiments/alberta_core_rl/configs/centered_td_diagnostics/config_main.json
```